

BST Physics Curriculum Vol 11 Chapter 9 — Cyclotomic $Q(\zeta_N)$ + Reed-Solomon v0.4 SIGNATURE (refilled per Cal #104)

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Vol 11 Chapter 9 — Cyclotomic $Q(\zeta_N)$ + Reed-Solomon

Chapter motivation

Cyclotomic field theory: $Q(\zeta_N)$ = field generated over rationals by N -th root of unity $\zeta_N = e^{2\pi i/N}$; Galois group $\text{Gal}(Q(\zeta_N)/Q) \cong (\mathbb{Z}/N\mathbb{Z})^*$; standard abstract algebra (Lang + Hungerford + Dummit-Foote). Kronecker-Weber theorem: every abelian extension of Q is contained in some $Q(\zeta_N)$.

Reed-Solomon error-correcting codes (Reed-Solomon 1960): error-correcting codes over finite field $GF(q)$ using polynomial evaluation at q field elements; can correct up to $(n - k)/2$ errors in a length- n codeword with k information symbols; widely used (CDs, DVDs, Blu-ray, QR codes, deep-space communication). Singleton bound: any (n, k) code has minimum distance $d \leq n - k + 1$; Reed-Solomon achieves Singleton bound (MDS codes).

BST cross-link: per K59 RATIFIED Tuesday + T2429 substrate-tick discretization, substrate operates on $GF(2^g) = GF(128)$ field per Koons tick ($g = 7$ BST primary Mersenne exponent; $2^7 - 1 = 127 = M_g$ Mersenne prime). Substrate-tick computation is 7-step cyclotomic projection chain on $GF(128)$ field extension; 7-step = 7 cyclotomic factors of $GF(128)$ over $GF(2)$. Reed-Solomon coding on $GF(128)$ is substrate-natural error-correcting framework for substrate-tick computation.

Section 9.0b — Reader-grade 3-level pedagogy

Level 1: Cyclotomic field theory $Q(\zeta_N)$ + Galois groups + Kronecker-Weber theorem + Reed-Solomon error-correcting codes on $GF(q)$; BST cross-link K59 RATIFIED + T2429: substrate-tick computation on $GF(2^g) = GF(128)$ field with 7-step cyclotomic projection chain.

Level 2 (graduate-mathematician): Standard cyclotomic field theory: $Q(\zeta_N) = Q[\zeta_N]$ field generated over rationals by N -th primitive root of unity $\zeta_N =$

$e^{2\pi i/N}$; $[Q(\zeta_N) : Q] = \varphi(N)$ Euler totient; Galois group $\text{Gal}(Q(\zeta_N)/Q) \cong (\mathbb{Z}/N\mathbb{Z})^*$ multiplicative group of integers mod N ; cyclotomic polynomial $\Phi_N(x)$ = minimal polynomial of ζ_N over Q . Kronecker-Weber theorem (Kronecker 1853 + Weber 1886): every finite abelian extension of Q is contained in some $Q(\zeta_N)$ — fundamental result in algebraic number theory + class field theory. Reed-Solomon error-correcting codes (Reed-Solomon 1960): codeword $(P(\alpha_1), P(\alpha_2), \dots, P(\alpha_n))$ where P is polynomial of degree $< k$ over $\text{GF}(q)$ + $\alpha_1, \dots, \alpha_n$ distinct elements of $\text{GF}(q)$; minimum distance $d = n - k + 1$ (Singleton bound achieved; MDS codes); decoding via syndrome computation + Berlekamp-Massey algorithm. Practical applications: CDs (Reed-Solomon RS(255, 223) over $\text{GF}(256)$), QR codes (Reed-Solomon over $\text{GF}(256)$), Voyager spacecraft communication, satellite TV. BST substrate cross-link: per K59 RATIFIED Tuesday 2026-05-19, substrate-tick computational framework is 7-step cyclotomic projection chain on $\text{GF}(2^g) = \text{GF}(128)$ where $g = 7$ BST primary (Mersenne exponent, $2^7 - 1 = 127 = M_g$ Mersenne prime). T2429 substrate-tick discretization (Lyra Thursday SP-31-1): per-tick Hilbert space = $\text{GF}(128)^k$ finite-dimensional; substrate-tick UV-completeness (Vol 1 Ch 10 T2437) inherits from finite-field $\text{GF}(128)$ structure. The 7 cyclotomic projection steps correspond to 7 cyclotomic-polynomial factors of $x^{128} - x = \prod_{d|7} \Phi_d(x)$ (modulo characteristic-2 reduction); $\text{GF}(128) = \text{GF}(2)[x]/(\Phi_7(x))$ cyclotomic-polynomial quotient. Reed-Solomon on $\text{GF}(128)$ substrate-natural error-correction: substrate uses Reed-Solomon-equivalent coding for substrate-tick state propagation (Paper #122 Information Substrate framing). Cyclotomic $Q(\zeta_{128})$ Galois group $\text{Gal}(Q(\zeta_{128})/Q) \cong (\mathbb{Z}/128\mathbb{Z})^*$ of order $\varphi(128) = 64 = 2^6$ (BST primary 6 squared) — substrate-coding Galois symmetry matches BST C_2 exactly. Cross-link Vol 1 Ch 10 + Vol 6 Ch 9 (RG flow = 7-step cyclotomic chain) + Vol 14 Ch 2 (Reed-Solomon Coding on $\text{GF}(128)$) for full substrate-coding theory development.

Level 3 (5th-grader accessible): Cyclotomic field theory studies the special complex numbers $e^{2\pi i/N}$ (N -th roots of unity). Reed-Solomon codes (used in CDs, QR codes, deep-space communication) are error-correcting codes using polynomials over finite fields like $\text{GF}(128)$. BST identifies the substrate's per-tick computation as a 7-step cyclotomic projection chain on $\text{GF}(128)$ — per K59 RATIFIED Tuesday + T2429. The 7 = BST integer g ; $128 = 2^7$. Substrate IS a Reed-Solomon-like coding framework per Casey CSE directive.

Section 9.1 — Cyclotomic Field Theory $Q(\zeta_N)$

$Q(\zeta_N)$ = field generated by N -th primitive root of unity; $[Q(\zeta_N) : Q] = \varphi(N)$; $\text{Gal}(Q(\zeta_N)/Q) \cong (\mathbb{Z}/N\mathbb{Z})^*$.

Cyclotomic polynomial $\Phi_N(x)$ = minimal polynomial of ζ_N over Q .

Section 9.2 — Kronecker-Weber Theorem

Every finite abelian extension of Q is contained in some $Q(\zeta_N)$. Fundamental result in algebraic number theory + class field theory.

Section 9.3 — Reed-Solomon Codes (1960)

Codeword $(P(\alpha_1), \dots, P(\alpha_n))$ for polynomial P of degree $< k$ over $\text{GF}(q)$.

Minimum distance $d = n - k + 1$ (Singleton bound; MDS codes).

Decoding: syndrome + Berlekamp-Massey algorithm.

Applications: CDs RS(255, 223) over $\text{GF}(256)$; QR codes; Voyager spacecraft; satellite TV.

Section 9.4 — BST K59 RATIFIED Substrate-Tick Cyclotomic Chain

Per K59 RATIFIED Tuesday 2026-05-19: substrate-tick computational framework is 7-step cyclotomic projection chain on $\text{GF}(2^g) = \text{GF}(128)$.

$g = 7$ BST primary (Mersenne exponent; $2^7 - 1 = 127 = M_g$ Mersenne prime).

7 cyclotomic projection steps correspond to 7 cyclotomic-polynomial factors of $x^{128} - x$.

Section 9.5 — T2429 Substrate-Tick Discretization (Vol 1 Ch 10)

Per Lyra Thursday SP-31-1 T2429: per-tick Hilbert space = $\text{GF}(128)^k$ finite-dimensional.

Substrate-tick UV-completeness (Vol 1 Ch 10 T2437) inherits from finite-field $\text{GF}(128)$ structure.

$\text{GF}(128) = \text{GF}(2)[x] / (\Phi_7(x))$ cyclotomic-polynomial quotient.

Section 9.6 — Galois Group Substrate-Symmetry

$\text{Gal}(\mathbb{Q}(\zeta_{128})/\mathbb{Q}) \cong (\mathbb{Z}/128\mathbb{Z})^*$ order $\varphi(128) = 64 = 2^6$ (BST primary 6 squared).

Substrate-coding Galois symmetry matches BST C_2 exactly.

Section 9.7 — Reed-Solomon Substrate-Natural Framework

Reed-Solomon on $\text{GF}(128)$ is substrate-natural error-correction: substrate uses Reed-Solomon-equivalent coding for substrate-tick state propagation (Paper #122 Information Substrate framing).

Cross-link Vol 14 Ch 2 Reed-Solomon Coding on $\text{GF}(128)$ full substrate-coding development.

Section 9.8 — Honest scope + Connection

- Standard cyclotomic theory + Reed-Solomon coding [?]
- K59 RATIFIED + T2429 substrate-tick $\text{GF}(128)$ framework [?]

- Galois symmetry $\varphi(128) = 64 = 2^{\wedge}C_2$ BST primary match
- Open scope: full substrate-coding theorem-grade development (multi-month)

Connection: - Vol 1 Ch 10 Renormalization + substrate-tick UV-completeness - Vol 6 Ch 9 Critical phenomena RG (7-step cyclotomic chain) - Vol 14 Ch 2 Reed-Solomon Coding on GF(128) + Ch 8 BST Coding Theory - Vol 11 Ch 10 Mersenne primes ($g = 7 + M_g = 127$ connection)

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